



4750 - Unveiling the Primordial Universe: A <1000 Msun Stellar Cluster at $z=6.6$

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSpec				
	3	G140M+G395M: Repl nA PA103 v8	NIRSpec MultiObject Spectroscopy	(2) macs0416_go4750_main_v6
	4	G140M+G395M: Repl nB PA103 v8	NIRSpec MultiObject Spectroscopy	(2) macs0416_go4750_main_v6
	5	G140M+G395M: Repl nC PA103 v8	NIRSpec MultiObject Spectroscopy	(3) macs0416_go4750_main_v6c

ABSTRACT

The formation of primordial stars and galaxies in the early universe has been a challenge for observational astronomers until recent advancements with JWST. Notably, Vanzella et al. (2023) used NIRSpec-IFU observations to study a highly magnified ($>\sim 100\times$) Lyman-alpha emitting arc at $z=6.6$, hinting at the presence of a low-mass ($M_{\text{star}} < 1000 M_{\text{sun}}$), low-metallicity stellar cluster within the arc. This suggests it might host

Population III stars formed from pristine gas.

This program proposes extensive NIRSpec spectroscopy of the stellar cluster. The main goal is to confirm the existence of primordial gas at $z=6.6$ and ongoing Population III star formation. To achieve this, deep observations requiring 24 hours of on-source integration will be conducted. These observations aim to identify the HeII 1640 emission line, which will serve as a critical test of the Population III star formation scenario. The objectives also include establishing a more stringent upper limit on gas-phase metallicity by examining the deep spectrum around [OIII] 5007, providing stronger support for the primordial nature within the star forming regions of the stellar cluster.

Furthermore, we will combine the entire suite of NIRCам images from GTO observations to obtain a stricter limit on the stellar mass of the stellar cluster, potentially down to 500 Msun. With these updated observational findings, this program will contribute to our understanding of the characteristics of Population III stellar clusters and the persistence of pristine gas conditions in the universe.

OBSERVING DESCRIPTION

This program conducts deep spectroscopy within a single pointing in the MACS J0416 lensing cluster field.

NIRSpec MSA MOS Spectroscopy:

The NIRSpec MSA MOS observations will be performed using the G140M/F070LP and G395M/F290LP gratings/filters, each requiring approximately 24 hours of on-source integration to achieve our desired depth.

A single MSA mask will be employed to simultaneously observe the two primary targets, B1 and B2, positioned close to each other, on the same MSA slitlet. This required a specific aperture position angle of 123-132.5 degrees, with an ideal APA of 126 degrees. This will ensure the other component, A1,2, falling on the slitlet. Within the same MSA mask, blank sky slitlets will be prepared to generate a master background spectrum. The background subtraction strategy, utilizing the master background spectrum for the primary targets, mitigates the issue of mutual subtraction and ensure a reliable background subtraction. Additionally, any remaining shutter positions will be utilized to observe other high-redshift lensed at MACS J0416.

Our observation strategy will employ a 3-shutter slitlet nod pattern, enhancing spectral sampling and the accuracy of flux measurements. To minimize the impact of 1/f noise in the master background, we will adopt the IRS2 readout mode.

No pre-imaging is required, as we will use the JWST NIRCам images that are taken by the GTO teams and will be publicly available.

Proposal 4750 - Targets - Unveiling the Primordial Universe: A <1000 Msun Stellar Cluster at z=6.6

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	macs0416_go4750_main_v6	RA: 04 16 9.7493 (64.0406221d) Dec: -24 04 31.91 (-24.07553d) Equinox: J2000		
	Comments: Description=[]				
Fixed Targets	(3)	macs0416_go4750_main_v6c	RA: 04 16 8.7610 (64.0365042d) Dec: -24 04 26.72 (-24.07409d) Equinox: J2000		
	Comments: Description=[]				

Proposal 4750 - Observation 3 - Unveiling the Primordial Universe: A <1000 Msun Stellar Cluster at z=6.6

Wed Oct 02 17:00:57 GMT 2024

Observation	Proposal 4750, Observation 3: G140M+G395M: ReplanA PA103 v8										Wed Oct 02 17:00:57 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: APA Change Request 89162 has been approved (on Jul 17, 2024; Japan time).										
	The primary targets are located in the quadrant Q2. Therefore, we kindly request that the alignment stars in Q2 be given the highest priority for the MSA alignment.										
	There are two designs, ca1p2 and ca2p2, which have minor but scientifically significant differences in the slitlet assignments. A total of 8 configurations are included, with each configuration requiring approximately 2.5 hours (on-source). This brings the total on-source observation time for this block to around 20 hours.										
	Given the long duration (~20 hours), we would appreciate it if the alignment could be checked during the observation block to ensure continued accuracy.										
Diagnostics	The design includes 8 reference stars for alignment, selected from the NIRCam images (with photometry based on F200W), which share the same astrometry as the science targets.										
	We have enabled the Master Background option for each MOS design and placed some open shutters in sky regions (without sources) to generate the master background spectrum for sources that may require it due to potential difficulties in using noded data for local background estimation. Approximately 20 background shutters have been positioned in each quadrant (except for Q4), with some shutters being occasionally connected. In particular, Q2 contains the main target clumps, which will benefit from the master background spectrum.										
	Note that we plan to apply both the master and local background subtraction methods during the data reduction process to ensure the best possible spectra. While Q4 contains few or no background shutters, this should not pose any issues, as the sources in Q4 will be reduced using the local background subtraction method.										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	# Name Target Coordinates Targ. Coord. Corrections Miscellaneous										
	(2) macs0416_go4750_main_v6 RA: 04 16 9.7493 (64.0406221d) Dec: -24 04 31.91 (-24.07553d) Equinox: J2000 Comments: Description=[]										
Acquisition	# Reference Star Bin Target Filter MSA Configuration Readout Pattern Groups/Int Integrations/Exp Total Integrations Total Exposure Time ETC Wkbk.Calc ID										
	1 Filter: CLEAR; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy] SAME CLEAR Auto Acq MSA Config NRSRAPID 3 1 4 171.788										
Template	TA Method Obtain Confirmation Images Science Aperture Primary Candidate List Filler Candidate List Spectral Overlap Map Spectral Overlap Threshold										
	MSATA No MSA Center Primary2_v6 (125 sources) Filler2_v6 (811 sources) jwst-nirspec-mr 1.5										

Proposal 4750 - Observation 3 - Unveiling the Primordial Universe: A <1000 Msun Stellar Cluster at z=6.6

Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	5	64.082435	-24.101844	21.99	1	15	64.079397	-24.079107	22.20	
	1	7	64.078963	-24.104214	22.67	1	37	64.044815	-24.069847	23.01	
	1	11	64.081530	-24.086339	23.72	1	40	64.032601	-24.077570	22.30	
	1	12	64.067952	-24.114730	22.97	1	43	64.035693	-24.065433	21.65	
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	ca1p2	3 Shutter Slitlet	64.057085416666 67 Degrees - 24.090961111111 085 Degrees	102.99328802014 925			3	6	8840.867
	2	1 (G140M/F070LP)	ca1p2	3 Shutter Slitlet	64.057085416666 67 Degrees - 24.090961111111 085 Degrees	102.99328802014 925			3	6	8840.867
	3	2 (G395M/F290LP)	ca1p2	3 Shutter Slitlet	64.057085416666 67 Degrees - 24.090961111111 085 Degrees	102.99328802014 925			3	6	8840.867
	4	2 (G395M/F290LP)	ca1p2	3 Shutter Slitlet	64.057085416666 67 Degrees - 24.090961111111 085 Degrees	102.99328802014 925			3	6	8840.867
	5	1 (G140M/F070LP)	ca2p2	3 Shutter Slitlet	64.057085416666 67 Degrees - 24.090961111111 085 Degrees	102.99328802014 925			3	6	8840.867
	6	1 (G140M/F070LP)	ca2p2	3 Shutter Slitlet	64.057085416666 67 Degrees - 24.090961111111 085 Degrees	102.99328802014 925			3	6	8840.867
	7	2 (G395M/F290LP)	ca2p2	3 Shutter Slitlet	64.057085416666 67 Degrees - 24.090961111111 085 Degrees	102.99328802014 925			3	6	8840.867
	8	2 (G395M/F290LP)	ca2p2	3 Shutter Slitlet	64.057085416666 67 Degrees - 24.090961111111 085 Degrees	102.99328802014 925			3	6	8840.867
Special Requirements	Aperture PA Range 103.00 to 103.00 Degrees (V3 324.4254303 to 324.4254303) [MSA Selected] MSA Scheduled Aperture PA 103.0000 to 103.0000 Degrees (V3 324.4254303 to 324.4254303)										

Proposal 4750 - Observation 4 - Unveiling the Primordial Universe: A <1000 Msun Stellar Cluster at z=6.6

Wed Oct 02 17:00:57 GMT 2024

Observation	Proposal 4750, Observation 4: G140M+G395M: ReplanB PA103 v8										Wed Oct 02 17:00:57 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: APA Change Request 89162 has been approved (on Jul 17, 2024; Japan time).										
	The primary targets are located in the quadrants Q2 and Q4. Therefore, we kindly request that the alignment stars in Q2 and Q4 be given the highest priority for the MSA alignment.										
	There are two designs, cb1p2 and cb2p2, which have minor but scientifically significant differences in the slitlet assignments. A total of 8 configurations are included, with each configuration requiring approximately 2.5 hours (on-source). This brings the total on-source observation time for this block to around 20 hours.										
	Given the long duration (~20 hours), we would appreciate it if the alignment could be checked during the observation block to ensure continued accuracy.										
Diagnostics	The design includes 8 reference stars for alignment, selected from the NIRCam images (with photometry based on F200W), which share the same astrometry as the science targets.										
	We have enabled the Master Background option for each MOS design and placed some open shutters in sky regions (without sources) to generate the master background spectrum for sources that may require it due to potential difficulties in using noded data for local background estimation. Approximately 20 background shutters have been positioned in each quadrant (except for Q4), with some shutters being occasionally connected. In particular, Q2 contains the main target clumps, which will benefit from the master background spectrum.										
	Note that we plan to apply both the master and local background subtraction methods during the data reduction process to ensure the best possible spectra. While Q4 contains few or no background shutters, this should not pose any issues, as the sources in Q4 will be reduced using the local background subtraction method.										
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	macs0416_go4750_main_v6	RA: 04 16 9.7493 (64.0406221d) Dec: -24 04 31.91 (-24.07553d) Equinox: J2000								
Acquisition	Comments: Description=[]										
	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
Template	1	Filter: CLEAR; Readout: NRSRAPIDD2; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD2	3	1	4	343.577	
	TA Method	Obtain Confirmation Images	Science Aperture	Primary Candidate List		Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
	MSATA	No	MSA Center	Primary2_v6 (125 sources)		Filler2_v6 (811 sources)		jwst-nirspec-mr		1.5	

Proposal 4750 - Observation 4 - Unveiling the Primordial Universe: A <1000 Msun Stellar Cluster at z=6.6

Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	7	64.078963	-24.104214	22.67	1	40	64.032601	-24.077570	22.30	
	1	18	64.069147	-24.085830	22.50	1	41	64.029006	-24.085616	22.47	
	1	20	64.061220	-24.096497	22.49	1	42	64.044259	-24.048955	23.67	
	1	39	64.047438	-24.050616	23.75	1	50	64.030041	-24.057472	23.05	
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	cb1p2	3 Shutter Slitlet	64.05808 Degrees - 24.070411111111 127 Degrees	102.99291128625 34			3	6	8840.867
	2	1 (G140M/F070LP)	cb1p2	3 Shutter Slitlet	64.05808 Degrees - 24.070411111111 127 Degrees	102.99291128625 34			3	6	8840.867
	3	2 (G395M/F290LP)	cb1p2	3 Shutter Slitlet	64.05808 Degrees - 24.070411111111 127 Degrees	102.99291128625 34			3	6	8840.867
	4	2 (G395M/F290LP)	cb1p2	3 Shutter Slitlet	64.05808 Degrees - 24.070411111111 127 Degrees	102.99291128625 34			3	6	8840.867
	5	1 (G140M/F070LP)	cb2p2	3 Shutter Slitlet	64.05808 Degrees - 24.070411111111 127 Degrees	102.99291128625 34			3	6	8840.867
	6	1 (G140M/F070LP)	cb2p2	3 Shutter Slitlet	64.05808 Degrees - 24.070411111111 127 Degrees	102.99291128625 34			3	6	8840.867
	7	2 (G395M/F290LP)	cb2p2	3 Shutter Slitlet	64.05808 Degrees - 24.070411111111 127 Degrees	102.99291128625 34			3	6	8840.867
	8	2 (G395M/F290LP)	cb2p2	3 Shutter Slitlet	64.05808 Degrees - 24.070411111111 127 Degrees	102.99291128625 34			3	6	8840.867
Special Requirements	Aperture PA Range 103.00 to 103.00 Degrees (V3 324.4254303 to 324.4254303) [MSA Selected] MSA Scheduled Aperture PA 103.0000 to 103.0000 Degrees (V3 324.4254303 to 324.4254303)										

Proposal 4750 - Observation 5 - Unveiling the Primordial Universe: A <1000 Msun Stellar Cluster at z=6.6

Wed Oct 02 17:00:57 GMT 2024

Observation	Proposal 4750, Observation 5: G140M+G395M: ReplanC PA103 v8										Wed Oct 02 17:00:57 GMT 2024	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Comments: APA Change Request 89162 has been approved (on Jul 17, 2024; Japan time).											
	The primary targets are located in the quadrant Q2. Therefore, we kindly request that the alignment stars in Q2 be given the highest priority for the MSA alignment.											
	There are a single design, cc1p2. A total of 4 configurations are included, with each configuration requiring approximately 2.5 hours (on-source). This brings the total on-source observation time for this block to around 10 hours.											
	Given the long duration (~10 hours), we would appreciate it if the alignment could be checked during the observation block to ensure continued accuracy. The design includes 8 reference stars for alignment, selected from the NIRCcam images (with photometry based on F200W), which share the same astrometry as the science targets. We have enabled the Master Background option for each MOS design and placed some open shutters in sky regions (without sources) to generate the master background spectrum for sources that may require it due to potential difficulties in using noded data for local background estimation. Approximately 20 background shutters have been positioned in each quadrant (except for Q4), with some shutters being occasionally connected. In particular, Q2 contains the main target clumps, which will benefit from the master background spectrum. Note that we plan to apply both the master and local background subtraction methods during the data reduction process to ensure the best possible spectra. While Q4 contains few or no background shutters, this should not pose any issues, as the sources in Q4 will be reduced using the local background subtraction method.											
Diagnosics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	macs0416_go4750_main_v6c	RA: 04 16 8.7610 (64.0365042d) Dec: -24 04 26.72 (-24.07409d) Equinox: J2000									
Comments: Description=[]												
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153		
Template	TA Method		Obtain Confirmation Images		Science Aperture		Primary Candidate List		Filler Candidate List		Spectral Overlap Map	Spectral Overlap Threshold
	MSATA		No		MSA Center		Primary2_v6c (125 sources)		Filler2_v6c (1288 sources)		jwst-nirspec-mr	1.5

Proposal 4750 - Observation 5 - Unveiling the Primordial Universe: A <1000 Msun Stellar Cluster at z=6.6

Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	14	64.075078	-24.095419	23.85	1	28	64.060098	-24.078894	24.81	
	1	16	64.068510	-24.093607	24.76	1	37	64.044815	-24.069847	23.01	
	1	20	64.061220	-24.096497	22.49	1	44	64.034492	-24.067992	22.93	
	1	22	64.058946	-24.094942	23.23	1	49	64.024026	-24.072958	24.73	
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	cc1p2	3 Shutter Slitlet	64.049364583333 34 Degrees - 24.082911111111 116 Degrees	102.99476167269 404			3	6	8840.867
	2	1 (G140M/F070LP)	cc1p2	3 Shutter Slitlet	64.049364583333 34 Degrees - 24.082911111111 116 Degrees	102.99476167269 404			3	6	8840.867
	3	2 (G395M/F290LP)	cc1p2	3 Shutter Slitlet	64.049364583333 34 Degrees - 24.082911111111 116 Degrees	102.99476167269 404			3	6	8840.867
	4	2 (G395M/F290LP)	cc1p2	3 Shutter Slitlet	64.049364583333 34 Degrees - 24.082911111111 116 Degrees	102.99476167269 404			3	6	8840.867
Special Requirements	Aperture PA Range 103.00 to 103.00 Degrees (V3 324.4254303 to 324.4254303) [MSA Selected] MSA Scheduled Aperture PA 103.0000 to 103.0000 Degrees (V3 324.4254303 to 324.4254303)										